

Nova X1 3D Printer - User Manual

Version 1.2. This manual covers setup, specifications, calibration, maintenance and troubleshooting for the Nova X1 FDM printer.

1. Technical Specifications

Parameter	Value
Build volume	220 x 220 x 250 mm
Nozzle diameter	0.4 mm (default), 0.2/0.6/0.8 optional
Max hotend temperature	300 C
Max bed temperature	110 C
Max print speed	300 mm/s
Filament diameter	1.75 mm
Extruder	Direct drive, dual gear
Bed	PEI spring steel, magnetic
Leveling	Automatic, 25-point mesh
Connectivity	USB-C, Wi-Fi, microSD

2. First-Time Setup

- Remove all foam and zip ties from the gantry.
- Attach the spool holder to the right side of the frame.
- Connect power and select your region's voltage (115V or 230V) on the PSU switch.
- Run Auto Level from Settings > Calibration. This takes about 4 minutes.
- Load filament: heat the nozzle to the material's print temperature, then push filament into the extruder until it flows.

3. Calibration

Run Z-offset calibration after every nozzle change. The default Z-offset is -1.20 mm. Adjust in 0.02 mm steps until the first layer is slightly squished but not transparent. Flow rate calibration should be repeated for each new filament brand.

4. Maintenance Schedule

Task	Interval	Notes
Clean PEI bed with IPA	Every 5 prints	Use 90%+ isopropyl alcohol
Lubricate Z lead screw	Every 200 hours	Use PTFE-based grease
Check belt tension	Every 100 hours	Belt should twang, not flop
Replace nozzle	Every 500 hours	Sooner with abrasive filaments
Clean extruder gears	Every 150 hours	Brush out filament dust
Firmware update	Monthly	Settings > About > Update

5. Troubleshooting

Problem	Likely cause	Fix
First layer not sticking	Z-offset too high or dirty bed	Lower Z-offset by 0.04 mm; clean bed with IPA
Stringing	Temperature too high or retraction too low	Lower temp 5 C; increase retraction to 1.0 mm
Clicking extruder	Partial clog or temp too low	Cold pull at 90 C (PLA); raise temp 5 C
Layer shifting	Loose belts or speed too high	Tighten belts; reduce speed 20%
Warping corners	Bed too cold or drafts	Raise bed temp 5 C; use enclosure for ABS
Error E03: Thermal runaway	Loose thermistor	Power off, reseal thermistor, contact support

6. Error Codes

E01 - Hotend heating failed. E02 - Bed heating failed. E03 - Thermal runaway detected. E04 - Filament runout. E05 - Auto-level probe failure. E06 - Wi-Fi module not detected.